

Simulation of crack patterns in quasi-brittle materials under thermal shock using phase field and cohesive zone models

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ABSTRACT

Complex crack patterns are usually formed on the surface of brittle and quasi-brittle solids under thermal shock. In this paper, the phase field coupled cohesive zone model is established to simulate the brittle and quasi-brittle fracture processes in three-dimensional solids, especially to reproduce the complex crack patterns under thermal shock. Several examples are given to illustrate the effectiveness of the model to complex fracture problems under thermal shock: from uniaxial tensile and thermal shrinkage fracture of a one-dimensional rod to nucleation, propagation, and crack pattern formation of thermal shock crack of a three-dimensional ceramic bottle. In addition, the formation mechanism of crack patterns in ceramic bottle is discussed, especially the dominant role of the difference in thermal expansion coefficients between the ceramic matrix layer and the glaze layer in the formation of crack patterns and the final crack density.

1. Introduction

Complex crack patterns are usually formed on the surface of brittle and quasi-brittle solids under thermal shock, e.g., hot ceramic tiles in cold water[1–3], thermal shock of α -alumina porous capillary[4,5], and instantaneous thermal start of engine blade[6]. Two typical thermal shock fracture patterns are shown in Fig. 1, including the ice crack in an incense-burner with fish-shaped handles of “Ge-kiln” porcelain of the Song Dynasty[7] and thermal shock crack patterns of a quenching ceramic plate[3]. Studying the formation mechanism of crack pattern under thermal shock is of great significance for the design and application of ceramic protective materials, as well as for the understanding and improvement of traditional ceramic technology.

Due to the complexity of the forming process of thermal shock crack patterns, numerical methods play an important role in fracture analyses. Researchers used boundary element method[8], non-local damage model[9], and damage mechanics-based model[10] to reproduce various crack patterns in quenching experiments. In addition, Menouillard and Belytschko[11] studied the thermal stress evolution and dynamic crack growth under thermal shock using the extended finite element method (XFEM). However, these numerical models still have limitations in simulating complex dynamic thermal shock-induced crack patterns. For such problems, there are usually dozens or even hundreds of cracks, and these cracks intersect, bifurcate, and merge with each other, which brings difficulties to their numerical solution. From a computational point of view, most of the aforementioned models suffer in situations with

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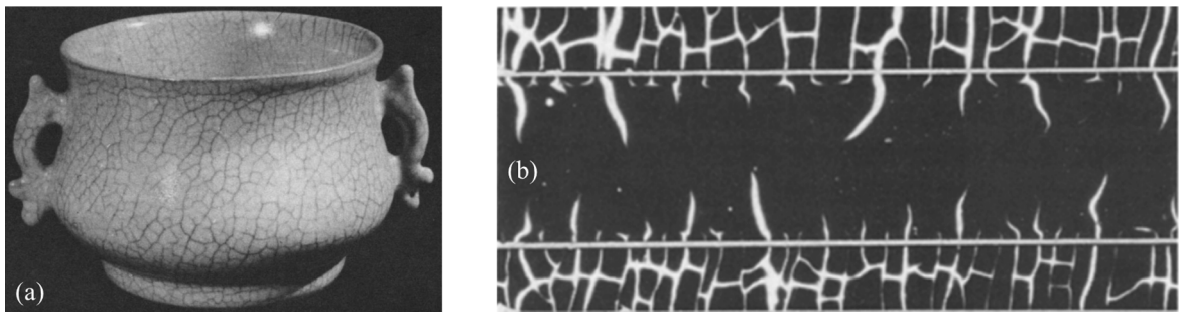


Fig. 1. Typical thermal shock fracture phenomenon: (a) ice crack in incense-burner with fish-shaped handles of “Ge-kiln” porcelain of Song Dynasty [7]; (b) Thermal-shock crack patterns of a quenching ceramic plate[3].

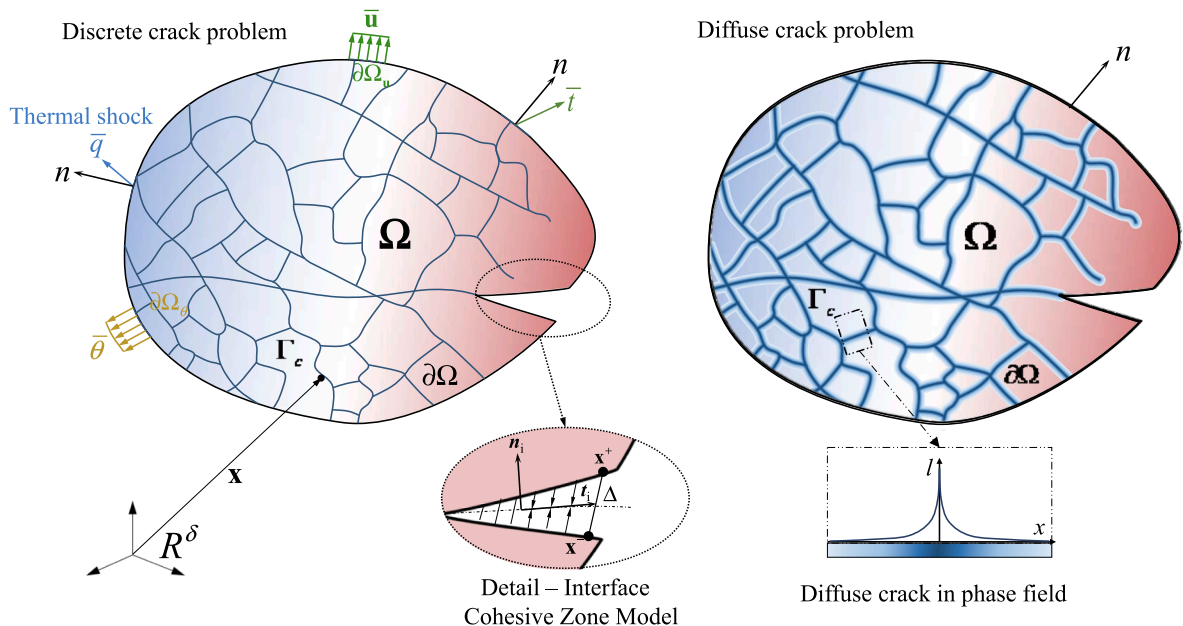


Fig. 2. Schematic diagram of an arbitrary solid with discontinuity in the domain under thermal shock load. Geometric regularization and phase field description of cracks.

complex crack topologies, providing substantial difficulties in numerical implementations, and often restrict the above formulations to simple crack topology[12].

In recent years, the phase field method has received much attention in modeling crack initiation, propagation, and complex crack networks[13–16]. Unlike other numerical methods such as XFEM[17,18], the phase field method does not need to introduce additional discontinuities. In contrast, the crack distribution is approximated by a field variable that smooths the crack surface in a small region. The main advantage of the phase field model is that the evolution of the crack surface follows the solution of the coupled partial differential equation (PDE). Since it is not necessary to follow the crack surface explicitly, the phase field approach is suitable for simulating complex crack patterns[19,20]. Therefore, this method has recently been extended to various complex fracture problems, including dynamic crack propagation and branching[21,22], composite failure[23,24], adiabatic shear fracture[25–27], hydrogen assisted fracture[28,29], hydraulic fracture[30–32], corrosion-induced cracking[33,34], fatigue fracture[35,36], blasting failure[37], and so on. It exhibits excellent prediction abilities, especially in multi-field coupled fracture problems[20,25,28,38].

Thermal shock coupled fracture is also a typical problem in the successful application of phase field model. Miehe et al.[15] developed a fully coupled thermo-mechanical phase field model for brittle fracture and studied the thermal shock-induced crack of glass ribbon and solid brick. Some researchers[20,39,40] further studied the thermal shock induced fracture of ceramic materials using phase field model, and captured the typical crack competition propagation mechanism and shielding effect. Li et al. [41] simulated the thermal shock of three-dimensional (3D) ceramic columns and considered the effect of temperature on material parameters. Although the above phase field simulations of thermal shock-induced fracture are in good agreement with the experimental results[1,42], there is no accurate simulation of surface complex crack patterns during quenching.

For the accurate simulation of the quenching crack pattern of ceramic materials, the key points include: (1) An accurate description

of the fracture failure process of quasi-brittle materials; (2) The transient inertia effect of thermal shock, convective heat transfer, and heat conduction process should be considered; (3) A fine enough mesh is needed to allow the initiation of a large number of initial damages. The results of conventional phase field fracture model (such as the AT1 and AT2 models[43]) generally have the problem of crack size sensitivity. The smaller the characteristic size parameter of the crack, the higher the ultimate bearing capacity of the structure[44]. To solve this problem, an effective approach is to combine the phase field model with the cohesive zone model. In this paper, the phase field coupled cohesive zone model is established to simulate the brittle and quasi-brittle fracture processes in 3D solids, especially to reproduce the complex crack patterns under thermal shock. In addition, the inertia effect of thermal shock, convective heat transfer, and transient heat conduction process are considered.

2. Thermo-elastic phase field approach combined cohesive zone model

This section outlines the basic aspects of the framework of the thermo-elastic phase field coupled with cohesive zone model. In particular, this method is derived by taking the formulas of Wu and Nguyen [45] as the basic results, which lays the foundation for combining the phase field method of overall fracture with the consistent interface formula relying on the classical cohesive zone model.

2.1. Variational formulation of thermo-elastic brittle fracture

The deformation and fracture of an arbitrary thermo-elastic solid in a general Euclidean space in the period $[0, t_a]$ are considered. The body occupies the domain $\Omega \in R^d$ represented by the boundary $\partial\Omega$, and its external normal unit vector is $\mathbf{n}$. As shown in Fig. 2, we assume that there is a discrete crack network Γ_c in the domain. The material points in the domain are identified by a separate position vector $\mathbf{x}$, while the displacement field is represented by $\mathbf{u}(\mathbf{x}, t)$ and the temperature field is represented by $\theta(\mathbf{x}, t)$. Finally, the body force $\mathbf{b}$, the boundary conditions in the form of specified displacement ($\bar{\mathbf{u}}$ on $\partial\Omega_u$) and specified traction ($\bar{\mathbf{t}}$ on $\partial\Omega_t$) act on the solid. In addition, the solid is affected by internal heat source r , and boundary conditions in the form of specified temperature ($\bar{\theta}$ on $\partial\Omega_\theta$) and specified heat flux ($\bar{\mathbf{q}}$ on $\partial\Omega_q$). These boundary conditions are subject to the following conditions:

$$\begin{aligned} \partial\Omega_t \cup \partial\Omega_u &= \partial\Omega, & \partial\Omega_t \cap \partial\Omega_u &= \emptyset \\ \partial\Omega_q \cup \partial\Omega_\theta &= \partial\Omega, & \partial\Omega_q \cap \partial\Omega_\theta &= \emptyset \end{aligned} \quad (1)$$

By establishing the variational fracture method as the starting point of the phase field model and formula, the thermal shock quasi-brittle material fracture considering the inertia effect is considered as a saddle point problem for finding [20]:

$$S[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] := \int_0^{t_a} \mathcal{L}[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] dt \quad (2)$$

where the Lagrangian $\mathcal{L}$ is defined as:

$$\begin{aligned} \mathcal{L}[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] := & \int_{\Omega/\Gamma_c} \left\{ \frac{1}{2} \rho \dot{\mathbf{u}} \cdot \dot{\mathbf{u}} - \psi_0(\boldsymbol{\epsilon}^e(\mathbf{u}, \theta)) + \rho \mathbf{b} \cdot \mathbf{u} + \rho c \dot{\theta} + \nabla \cdot \mathbf{q} - r \right\} dV \\ & + \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} d\Gamma_c + \int_{\partial\Omega_q} \bar{\mathbf{q}} \cdot \mathbf{n} d\Gamma_c - g_c |\Gamma_c| \end{aligned} \quad (3)$$

where ρ is the material density, c is the specific heat capacity, g_c is the critical energy release rate, and ψ_0 is the elastic strain energy density which depends on elastic strain:

$$\boldsymbol{\epsilon}^e = \boldsymbol{\epsilon} - \boldsymbol{\epsilon}^\theta \quad (4)$$

where $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}^\theta$ are total strain and thermal expansion strain, respectively, which are defined as:

$$\boldsymbol{\epsilon} := \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \quad \boldsymbol{\epsilon}^\theta := \alpha_\theta \cdot \Delta \mathbf{I} \quad (5)$$

where α_θ is the coefficient of thermal expansion (CTE), and $\mathbf{I}$ is the identity tensor.

In the phase field model, a damage variable $d: \Omega \rightarrow [0, 1]$ is introduced to describe the fracture and failure of the material. The material points with $d = 0$ and $d = 1$ are in unbroken and fully broken states, respectively. Therefore, the crack surface is dispersed, as shown in Fig. 2. According to the image segmentation theory, the energy contribution of the crack surface is obtained by the volume approximation with the following expression:

$$\int_{\Gamma_c} g_c dA \approx \int_{\Omega/\Gamma_c} g_c \cdot \gamma(d; \nabla d) dV \quad (6)$$

according to the unified phase field damage theory[14], crack surface density $\gamma(d; \nabla d)$ is the function of phase field variable d and its spatial gradient ∇d :

$$\gamma(d; \nabla d) = \frac{1}{c_a} \left[\frac{1}{l_c} \alpha(d) + l_c |\nabla d|^2 \right] \quad \text{with} \quad c_a = 4 \int_0^1 \sqrt{\alpha(\beta)} d\beta \quad (7)$$

where $l_c > 0$ is the length scale parameter, and the crack geometry function $\alpha(d) \in [0, 1]$ is an increasing function of the phase field

variable d , and satisfies the following conditions:

$$\alpha'(d) \geq 0, \quad \alpha(0) = 0, \quad \alpha(1) = 1 \quad (8)$$

then the regularized variational formulation reads: find $(\mathbf{u}, \theta, d)$ that is the stationary point of $S_{l_c}[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] := \int_0^{t_a} \mathcal{L}_{l_c}[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] dt$, with.

$$\begin{aligned} \mathcal{L}_{l_c}[\mathbf{u}, \dot{\mathbf{u}}, \theta, \Gamma_c] := & \int_{\Omega} \left\{ \frac{1}{2} \rho \dot{\mathbf{u}} \cdot \dot{\mathbf{u}} - \psi(\boldsymbol{\varepsilon}^e(\mathbf{u}, \theta), d) + \rho \mathbf{b} \cdot \mathbf{u} + \rho c \dot{\theta} + \nabla \cdot \mathbf{q} - r \right\} dV \\ & + \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} d\Gamma_c + \int_{\partial\Omega_q} \bar{\mathbf{q}} \cdot \mathbf{n} d\Gamma_c - g_c \int_{\Omega} \frac{1}{l_c} \left[\frac{1}{l_c} \alpha(d) + l_c |\nabla d|^2 \right] dV \end{aligned} \quad (9)$$

The strong form of the boundary value problem can be derived from Eq. (9):

$$\begin{aligned} \operatorname{div} \cdot \boldsymbol{\sigma} + \rho \mathbf{b} &= \rho \ddot{\mathbf{u}} & \text{in } \Omega \times [0, t_a] \\ \rho c \dot{\theta} + \nabla \cdot \mathbf{q} &= r & \text{in } \Omega \times [0, t_a] \\ -\frac{\partial \psi}{\partial d} - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] &= 0 & \text{in } \Omega \times [0, t_a] \end{aligned} \quad (10)$$

where $\boldsymbol{\sigma} = \partial \psi / \partial \boldsymbol{\varepsilon}^e$ is Cauchy stress. And the boundary conditions are as follows:

$$\begin{aligned} \boldsymbol{\sigma} \cdot \mathbf{n} &= \bar{\mathbf{t}} & \text{on } \partial\Omega_t \times [0, t_a] \\ \mathbf{q} \cdot \mathbf{n} &= \bar{\mathbf{q}} & \text{on } \partial\Omega_q \times [0, t_a] \\ \frac{\partial d}{\partial \mathbf{n}} &= 0 & \text{on } \partial\Omega \times [0, t_a] \end{aligned} \quad (11)$$

$$\begin{aligned} \mathbf{u} &= \bar{\mathbf{u}}, & \text{on } \partial\Omega_u \times [0, t_a] \\ \theta &= \bar{\theta}, & \text{on } \partial\Omega_\theta \times [0, t_a] \end{aligned} \quad (12)$$

where $\mathbf{q}$ is the heat flux, which is proportional to the temperature gradient:

$$\mathbf{q} = -k \cdot \nabla \theta \quad (13)$$

where k is the thermal conductivity of the material.

In the above equations, Eq. (11) is the second type of boundary condition (Neumann boundary condition), and Eq. (12) is the first type of boundary condition (Dirichlet boundary condition). The surface convective heat transfer boundary is critical to the quenching process, and its convective heat transfer is directly proportional to the temperature difference between the surface and the environment:

$$\mathbf{q} \cdot \mathbf{n} = -k \nabla \theta \cdot \mathbf{n} = \bar{h}(\theta_s - \theta_\infty) \quad \text{on } \partial\Omega_q \times [0, t_a] \quad (14)$$

where $\bar{h}$ is the surface convective heat transfer coefficient, θ_s is the surface temperature, and θ_∞ is the ambient temperature.

2.2. Phase-field regularized cohesive zone model (PF-CZM)

Considering the thermal shock fracture process of quasi-brittle materials, the free energy density function $\psi(\boldsymbol{\varepsilon}, \theta, d)$ defined as the energy degradation function $\omega(d)$ isotropic degenerate undamaged strain energy density function $\psi_0(\boldsymbol{\varepsilon}, \theta)$, namely:

$$\psi(\boldsymbol{\varepsilon}, \theta, d) = \omega(d) \psi_0(\boldsymbol{\varepsilon}, \theta) \quad (15)$$

with.

$$\psi_0(\boldsymbol{\varepsilon}, \theta) = \frac{1}{2} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\theta) : \mathbb{E}_0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^\theta) = \frac{1}{2} \bar{\boldsymbol{\sigma}} : \mathbb{C}_0 : \bar{\boldsymbol{\sigma}} \quad (16)$$

where, $\mathbb{E}_0$ and $\mathbb{C}_0$ are the fourth-order elastic stiffness and flexibility tensor of the material, respectively. $\bar{\boldsymbol{\sigma}}$ is the effective stress tensor, given by the following linear relation:

$$\bar{\boldsymbol{\sigma}} = \mathbb{E}_0 : \boldsymbol{\varepsilon}^e \quad (17)$$

The energy release rate of fracture phase field is defined as:

$$Y := -\frac{\partial \psi}{\partial d} = -\omega'(d) \mathcal{H} \quad (18)$$

in which, $\mathcal{H} := \partial \psi / \partial \omega$ is the effective energy release rate, it is also the driving force of phase field evolution. $\omega'(d) := \partial \omega / \partial d$ is the derivative of the energy degradation function.

To consider the different mechanical behaviors under tensile and compressive stress states, the uniaxial tensile strength was taken

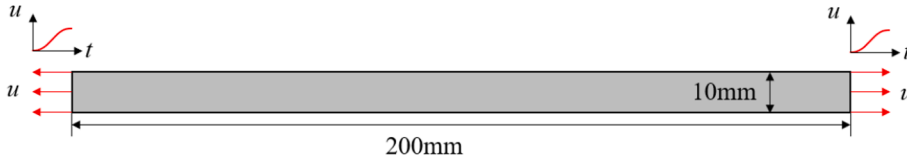


Fig. 3. Uniaxial tensile cracking of rectangular section bars: geometric and boundary conditions.

as the reference and the effective energy release rate $\mathcal{H}$ was redefined as [45]:

$$\mathcal{H} = \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2 \quad (19)$$

where, $\bar{\sigma}_1$ is the maximum principal stress of the effective stress tensor $\bar{\sigma}$. E_0 is the Young's modulus of the material. $\langle a \rangle_\pm := (a \pm |a|)/2$ for any $a \in \mathbb{R}$. Then the governing equation of phase field variable in Eq. (10) can be written as:

$$-\omega'(d)H - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] = 0 \quad (20)$$

To adopt explicit time integral, we further replace Eq. (20) with adding a time-dependent term according to Ref. [46,47]:

$$\dot{d} = \begin{cases} \frac{1}{\eta} \left\langle -\omega'(d)\mathcal{H} - g_c \frac{1}{c_a} \left[\frac{1}{l_c} \alpha'(d) - 2l_c \Delta d \right] \right\rangle_+, & d < 1 \\ 0, & \text{otherwise} \end{cases} \quad (21)$$

where η is the viscous parameter that governs phase field evolution.

To ensure that cracks do not heal, $\mathcal{H}$ needs to meet the following conditions:

$$\mathcal{H} = \begin{cases} \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2, & \text{if } \frac{1}{2E_0} \langle \bar{\sigma}_1 \rangle_+^2 > \mathcal{H}_n \\ \mathcal{H}_n, & \text{otherwise} \end{cases} \quad (22)$$

where $\mathcal{H}_n$ is the previously calculated history variable in the n -th increment step.

To better solve the problems of crack initiation and quasi-brittle failure, in accordance with the formulas of Wu [14], the following generic expressions are considered:

$$\alpha(d) = 2d - d^2, \quad \omega(d) = \frac{(1-d)^2}{(1-d)^2 + a_1 d \cdot (1-0.5d)}, \quad c_a = \pi \quad (23)$$

where.

$$a_1 = \frac{4}{\pi} \cdot \frac{l_{ch}}{l_c}, \quad l_{ch} = E_0 g_c / f_t^2 \quad (24)$$

where f_t is the tensile failure strength, and l_{ch} is the Griffith or Irwin length that characterizes the brittleness of solid material. The smaller the value of l_{ch} , the more brittle the material is.

The above model is a phase field regularized cohesive zone model considering inertia effect and thermal-mechanical coupling, which is suitable for the complex failure and fracture of quasi-brittle materials under quasi-static, dynamic and thermal shock loads. By writing user subroutines VUMAT and VHETVAL, and combining them with other subroutines including VUFIELD and VUSDFLD, the above PF-CZM is implemented in the commercial finite element package Abaqus/Explicit.

3. Numerical examples

3.1. Uniaxial tensile cracking of a rectangular section bar

In this section, the fracture of a rectangular section bar (1D bar) with length of $L = 200$ mm, height of $H = 10$ mm and thickness of $b = 1$ mm under uniaxial tension is considered, as shown in Fig. 3. To simulate the pure one-dimensional (1D) loading condition, the displacement boundaries in the $X \pm$ direction respectively are applied at both ends of the rod without constraining the degrees of freedom (Dofs) in other directions. The displacement boundaries are applied by a smooth amplitude curve. The maximum displacement applied at both ends is 0.05 mm. We controlled crack initiation in the middle part of the rod by introducing a small defect at that location to facilitate the comparison of the results under different parameters. Specifically, this was done by shifting the coordinates of a node at the top of the middle location of the rod inward by 1 % of the side length of an element. The material and length scale parameters of the bar are shown in Table 1. Unless otherwise specified, the viscosity parameter η in this paper are taken as $1.0e^{-6} \text{ kN}\cdot\text{s}/\text{mm}^2$.

The phase field profiles at the displacement of 0.05 mm (the extension of the whole rod is 0.1 mm) are shown in Fig. 4. As expected,

Table 1
Material and length scale parameters used in this subsection.

Parameter	E	ν	g_c	f_t	l_c
Value	30	0.2	0.1	3.0	2.5 ~ 10
Unit	GPa	–	N/mm	MPa	mm



Fig. 4. Phase field value distribution of uniaxial tensile fracture of a rectangular section bar simulated by regularized PF-CZM: (a) $l_c = 10$ mm, $h = 1$ mm; (b) $l_c = 5$ mm, $h = 0.5$ mm; (c) $l_c = 2.5$ mm, $h = 0.5$ mm; where h is the mesh size.

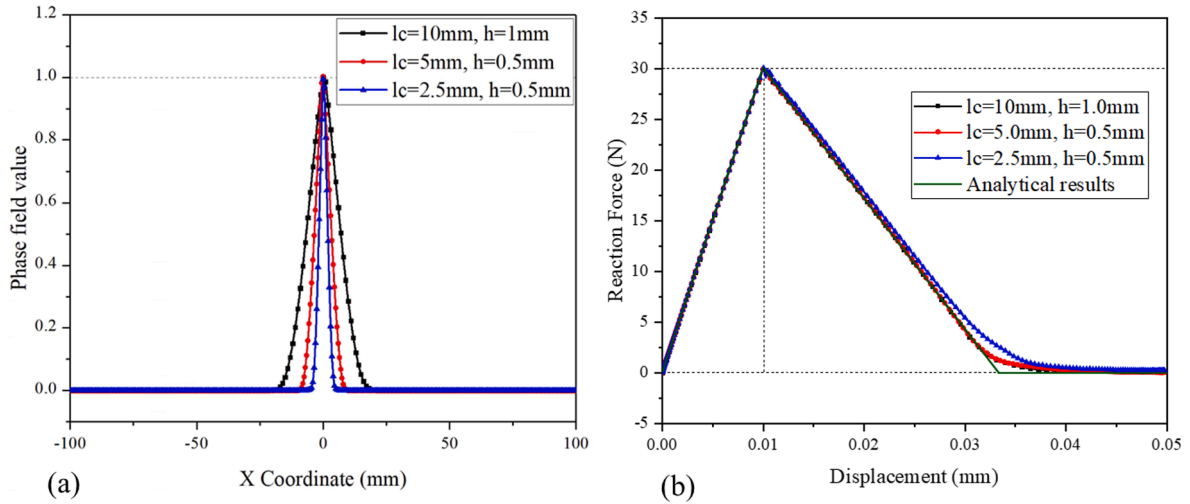


Fig. 5. (a) Distribution of phase field value along the rod axis and (b) the curve of reaction force versus displacement simulated by PF-CZM with different mesh sizes and length scale parameters.

the crack bandwidth decreases as the length scale parameter l_c becomes smaller, which can also be observed in Fig. 5(a). The numerical results of reaction force versus displacement under different length scale parameters and analytical results[48] are shown in Fig. 5(b), which is the reaction force–displacement curve of a typical bilinear cohesion model. It is noted that the nonlinearity when the load degenerates to near zero is due to the system viscosity introduced by the solver. Fig. 5 also shows that the length scale parameter does not affect the overall response of the structure, which is in sharp contrast to the existing brittle fracture phase field damage models. This is also one of the advantages of PF-CZM.

The curve of elastic energy, fracture energy, and external work versus displacement for the uniaxial tensile of 1D bar simulated by PF-CZM under different mesh sizes and length scale parameters are shown in Fig. 6. Firstly, the external work causes the increase of elastic strain energy. When the tensile stress reaches the rod's strength, the fracture energy begins to increase, and the elastic strain energy gradually decreases. Finally, all the external work is transformed into fracture energy, and the structure is completely broken.

The critical fracture energy is an important factor affecting the cohesive zone model behavior. The uniaxial tensile and fracture behaviors of the rectangular bar with different fracture energies were studied with other parameters unchanged. Fig. 7 shows the load–displacement curve of the rod under uniaxial tension under three critical fracture energy parameters and the analytical results [48], all simulation results showing good bilinear behavior. The smaller the g_c , the more brittle the material is.

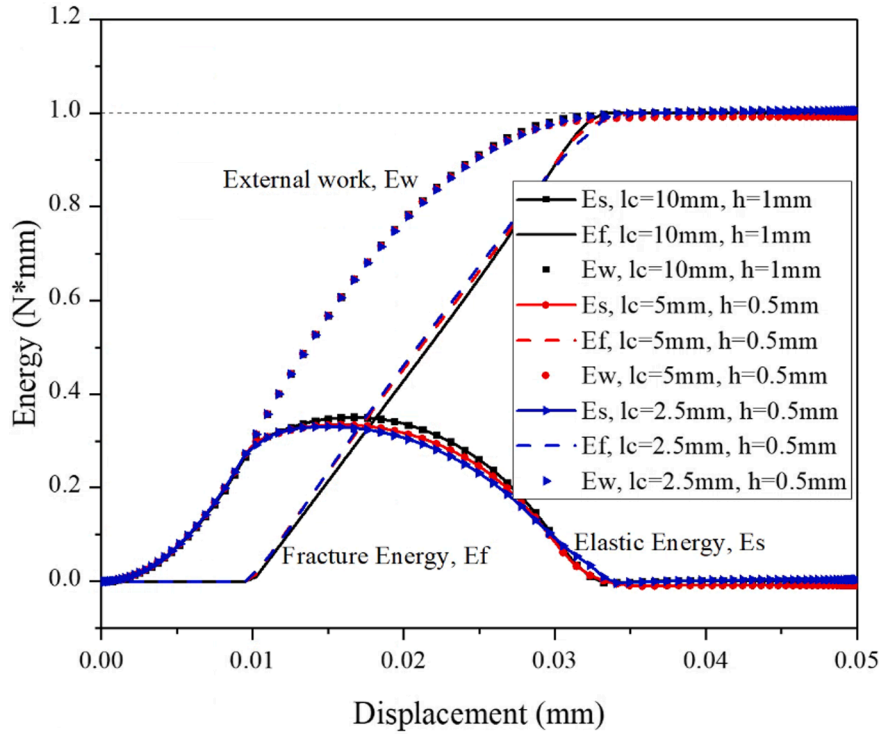


Fig. 6. Uniaxial tensile cracking of rectangular section bar: the curve of elastic energy, fracture energy and external work versus displacement simulated by PF-CZM with different mesh sizes and length scale parameters.

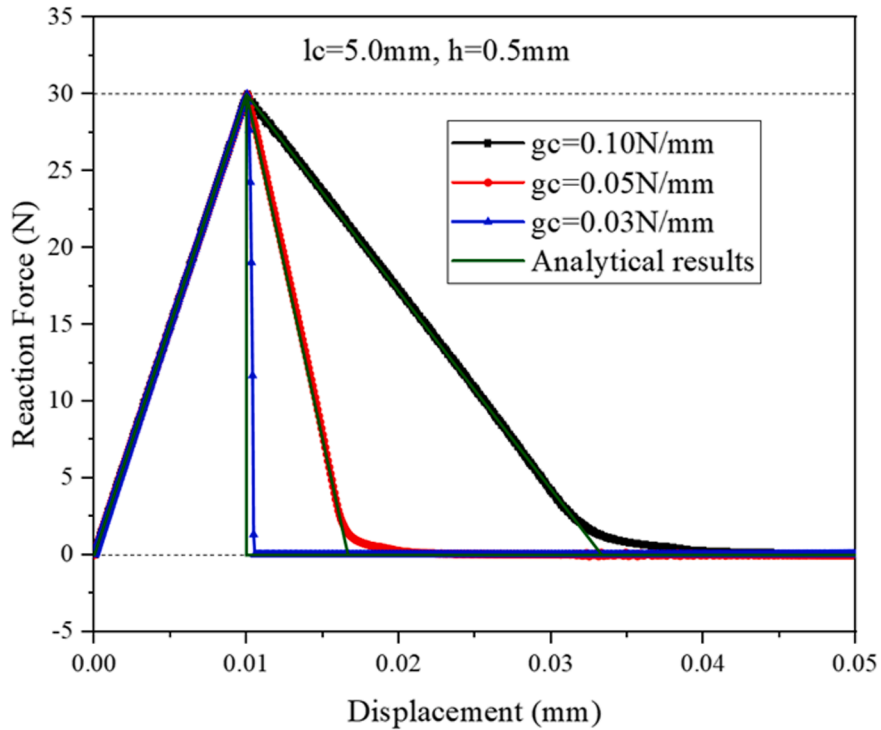


Fig. 7. Uniaxial tensile fracture of rectangular bar simulated by PF-CZM: Reaction force–displacement curves with different critical fracture energies g_c .

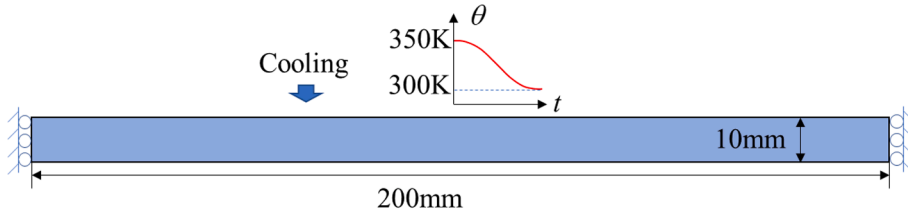


Fig. 8. Cold shrinkage cracking of rectangular section bars: geometric and boundary conditions.

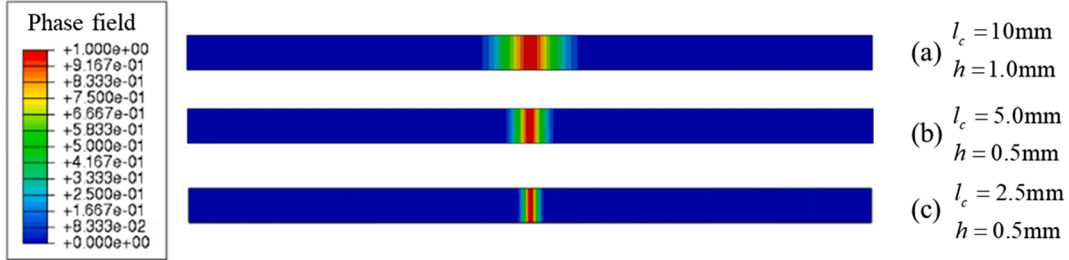


Fig. 9. Phase field value distribution of cold shrinkage cracking of a rectangular section bar simulated by regularized PF-CZM: (a) $l_c = 10\text{ mm}$, $h = 1\text{ mm}$; (b) $l_c = 5\text{ mm}$, $h = 0.5\text{ mm}$; (c) $l_c = 2.5\text{ mm}$, $h = 0.5\text{ mm}$.

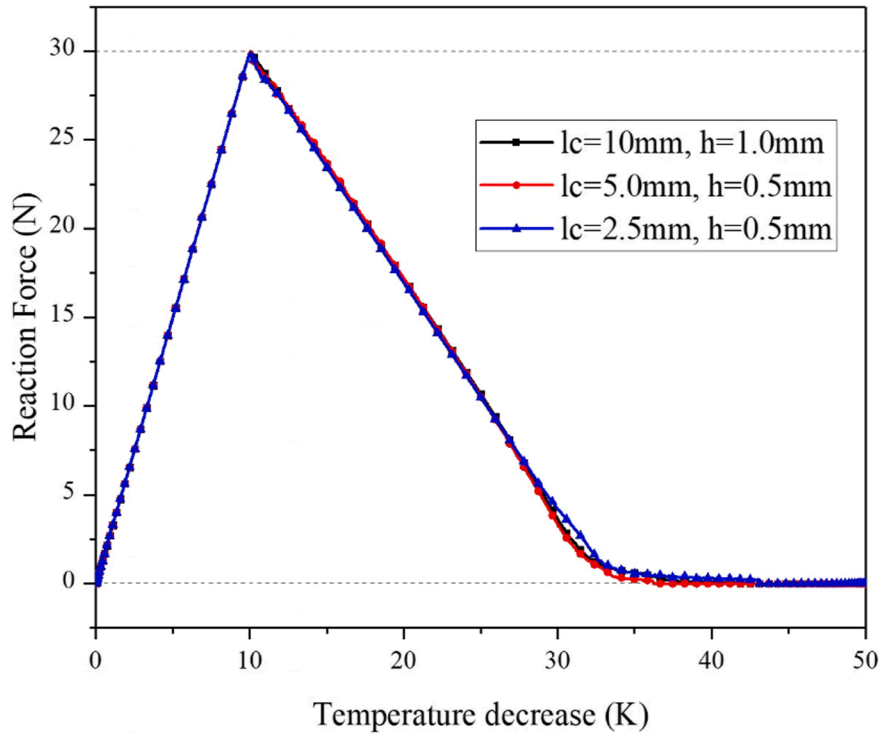


Fig. 10. Cold shrinkage cracking of a rectangular section bar: the curve of reaction force versus temperature decrease simulated by PF-CZM with different mesh sizes and length scale parameters.

3.2. Cold shrinkage cracking of rectangular section bar

To verify the fracture process under thermal–mechanical coupling, we consider the cold shrinkage cracking of a 1D rectangular section bar. Both ends of the rod are fixed. The geometric and material parameters are the same as those in the previous section. Besides, the CTE of the bar is $\alpha_0 = 1.0 \times 10^{-5}/\text{K}$. The initial temperature of the bar is 350 K, which is slowly reduced to 300 K by a smooth

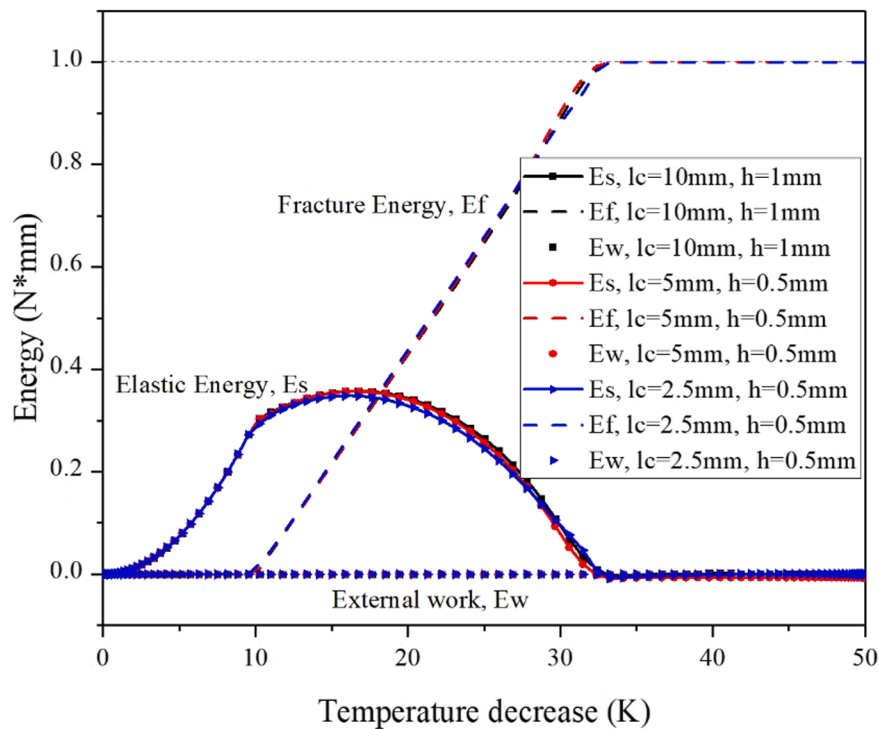


Fig. 11. Cold shrinkage cracking of a rectangular section bar: the curve of elastic energy, fracture energy and external work versus temperature decrease simulated by PF-CZM with different mesh sizes and length scale parameters.

Table 2

Material parameters of phase field simulation for ceramic plate quenching.

Parameter	E	ν	g_c	f_t	ρ	c	k_0	α_θ
Value	370	0.22	0.0425	180	2450	840	20	6.0e-6
Unit	GPa	–	N/mm	MPa	kg/m ³	J/kg*K	W/m*K	1/K

amplitude curve, as shown in Fig. 8.

The phase field value distributions under different mesh sizes and length scale parameters are shown in Fig. 9 as the rod temperature decreases by 50 K. Fig. 10 shows the curve of the unilateral load of the structure with temperature decrease under different mesh sizes and length scale parameters. Consistent with the conclusions in the previous section, the fracture response of the structure is independent of the mesh size and length scale parameters when considering the thermal-mechanical coupling.

The curve of elastic energy, fracture energy, and external work versus temperature decrease for the uniaxial tensile of 1D bar simulated by PF-CZM under different mesh sizes and length scale parameters are shown in Fig. 11. Initially, the elastic strain energy gradually increases as the temperature decreases. When the thermal stress in the rod exceeds the tensile strength, the phase field begins to evolve, the fracture energy gradually increases. Then, the structure begins to break, and the elastic strain energy gradually decreases to zero. During the whole process, the external force does not do any work, and the source of energy comes entirely from thermal energy.

3.3. Three-dimensional quenching cracking of ceramic plate

This example concerns a quenching test to investigate the thermo-elastic coupled crack process under thermal shock loading. Bahr et al. [3], Shao et al. [1] and Xu et al. [49] carried out detailed experimental studies on the quenching process of ceramic plates. The ceramic plates were heated to different temperatures and then put into a low-temperature water bath in the experiment. Under the thermal shock, many parallel cracks were produced on the side surface the ceramic, while interlaced crack pattern were produced on the cooled surface. In the quenching process, the shrinkage of ceramics with higher temperature after contact with water was more intense, showing a larger number of parallel cracks with alternating lengths. Many researchers have studied this problem numerically [20,39,41]. However, most of them are carried out in two-dimensional (2D) condition, without considering the process of crack pattern formation on the cooled surface.

We carried out the numerical simulation according to the experimental settings. The size of the sample is $60 \times 10 \times 6$ mm. To

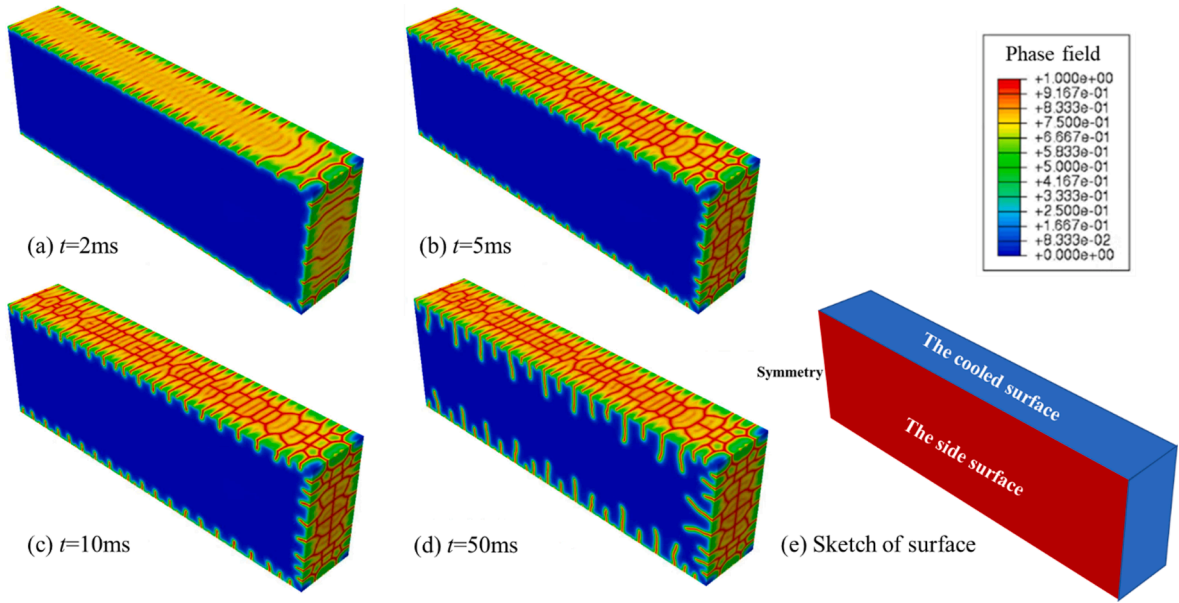


Fig. 12. Simulated crack phase field distribution of a ceramic plate under thermal shock at different moments: (a) $t = 2$ ms; (b) $t = 5$ ms; (c) $t = 10$ ms; (d) $t = 50$ ms. (e) Schematic diagram of the cooled surface and the side surface.

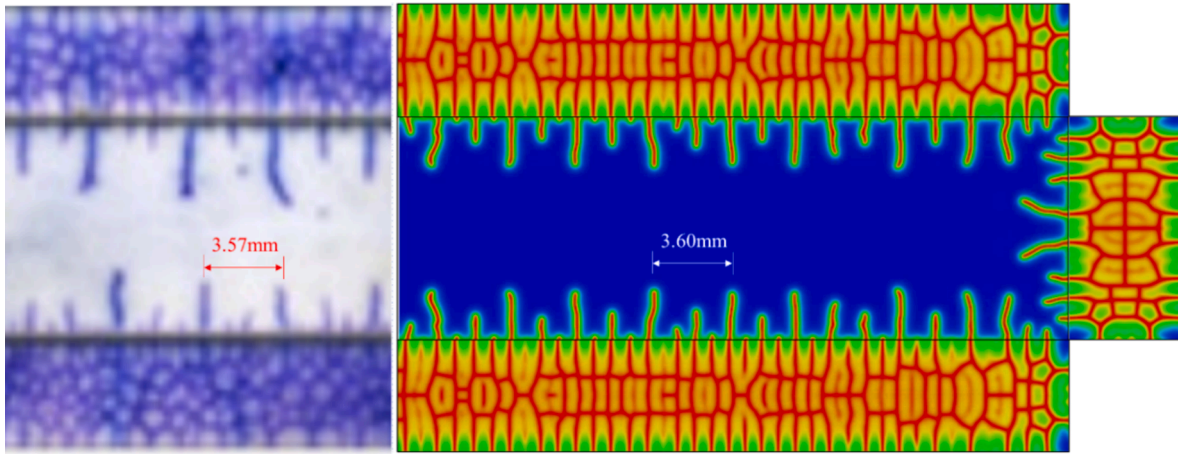


Fig. 13. Phase field simulation results of thermal shock crack of ceramic plates: surface phase field distribution under three views at 50 ms (right), as well as the comparison with previous experimental results (left) [49].

improve the computational efficiency, we take the $1/8$ model to simulate. The ambient temperature is $\theta_m = 300$ K, and the initial temperature of the plate is $\theta_i = 600$ K (the corresponding temperature difference is $\Delta\theta = 300$ K). The convective heat transfer coefficient of ceramic plate surface and water bath environment is $h = 55$ N/(mm $\cdot$ s $\cdot$ K). The material properties are listed in Table 2. To ensure the crack initiation process is accurately described at the boundary, the mesh size h is set to 0.05 mm (the total number of elements is 3,600,000) and the length scale parameter $l_c = 2h$. The parameter l_c ensures that enough microcracks are initiated to simulate subsequent competitive arresting and growth processes accurately. In this case, the crack initiation or propagation is mainly caused by thermal expansion and contraction.

The crack phase field distribution of ceramic plate under thermal shock at different moments are given in Fig. 12. The cooled surface and side surface of the ceramic plate are also shown in Fig. 12(e). If short cracks are neglected, the crack patterns calculated by numerical method at different moments are in good agreement with the experimental results. In fact, shorter cracks may nucleate in the ceramic slab during the experiment, but due to the limitations of the experimental technology, they cannot be clearly observed. Our PF-CZM simulations can well capture the two key features of this quenching process: (a) on the side faces, many parallel short cracks nucleate, then competitive propagate and arrest; (b) on the cooled surface, isotropic shrinkage leads to a large number of interlaced cracks on the surface, forming a cracking pattern.

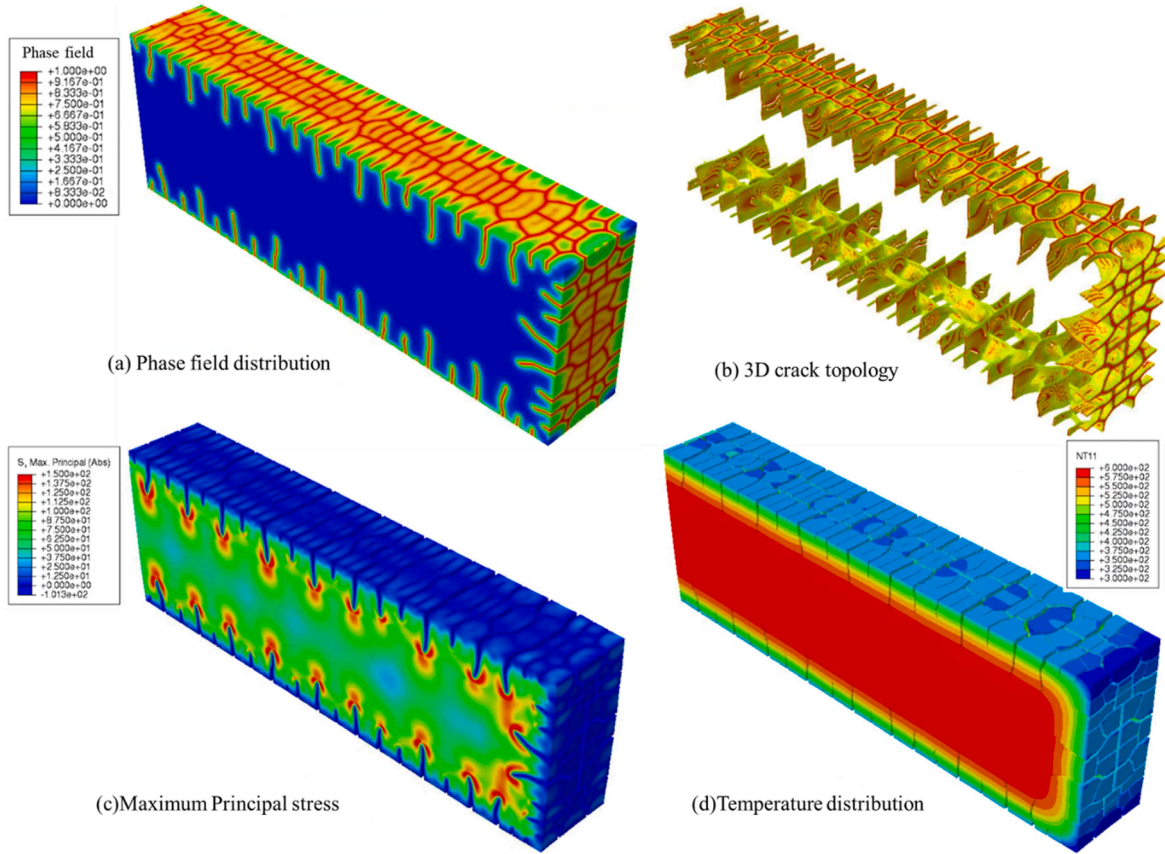


Fig. 14. Phase field simulation results of thermal shock fracture of ceramic plates at 50 ms: (a) phase field distribution; (b) 3D crack topology; (c) maximum principal stress distribution; (d) temperature distribution. The deformation scale factor in (c) and (d) is 20.

Fig. 13 shows the simulation results under three views and the comparison with the two views of previous experimental results[49] (there are only two views in the literature). The simulation results are in good agreement with the previous experimental results, including the distribution spacing of long cracks, crack density, etc. It should be pointed out that the slight difference between the experimental and numerical results is mainly because the experimental results are fully quenching, while the simulation results are only calculated to 50 ms due to the calculation cost.

The complex response behaviors of 3D ceramic plate during quenching are shown in Fig. 14, including 3D distribution of phase field, 3D geometric topology of crack space, distribution of maximum principal stress and temperature field.

3.4. Formation of the surface crack pattern of ceramic bottle

The reticular opening on the glaze is the most significant feature of “Ge-kiln” porcelain. The coarser cracks and the finer cracks are intertwined. The asymmetrical geometry leads to the formation of complex mixed cracks on the surface[50]. The mechanism of its formation in firing and quenching has always been a matter of concern. In this section, we use the PF-CZM to simulate the quenching process of a typical ceramic bottle, reproduce the formation process of its complex crack pattern, and explain some typical phenomena in its formation.

The ceramic bottle is a rotationally symmetrical structure. The sectional shape, geometric dimension parameters and local mesh details are shown in Fig. 15. The thickness of the outer glaze layer of the ceramic bottle is 0.8 mm, and the thickness of the inner matrix layer is 1.2 mm. To reduce the calculation cost, we use a 1/4 model for the simulation. The material parameters are the same as the ceramic plates in the previous section except for the CTE. The CTE of the ceramic inner matrix layer is $\alpha_1 = 5 \times 10^{-6}/K$, the CTE of the outer glaze layer is $\alpha_2 = 6 \times 10^{-6}/K$. So, the CTE difference is $\Delta\alpha = 1 \times 10^{-6}/K$. The initial temperature of the ceramic bottle was 593 K, and it was suddenly put into an environment with a temperature of 293 K. The convective heat transfer coefficient between the outer surface of the ceramic bottle and the environment is $\bar{h} = 55 N/(mm \cdot s \cdot K)$.

The phase field simulation results of crack propagation and crack pattern formation on the surface of the ceramic bottle under rapid cooling at different moments are shown in Fig. 16. It can be seen that a stable crack pattern is formed on the surface of the ceramic bottle within 10 ms. After 10 ms, the crack pattern basically does not change. Fig. 17 shows the 3D phase field distribution, 3D cracks geometric topology, temperature field distribution and principal stress distribution of the ceramic bottle at 10 ms. To observe the effect

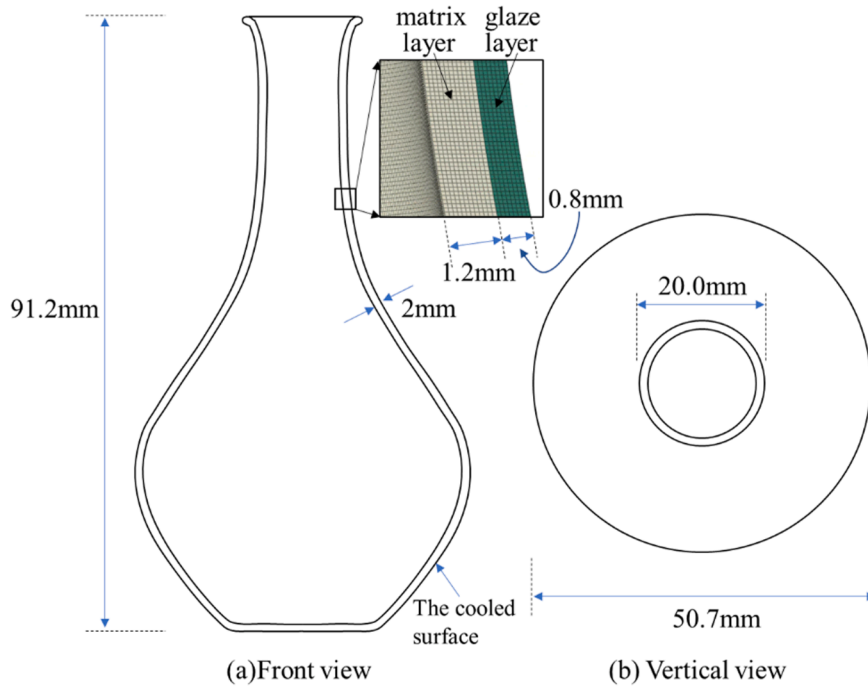


Fig. 15. Schematic diagram of cross-section shape, geometric dimension parameters, and local mesh details of the ceramic bottle.

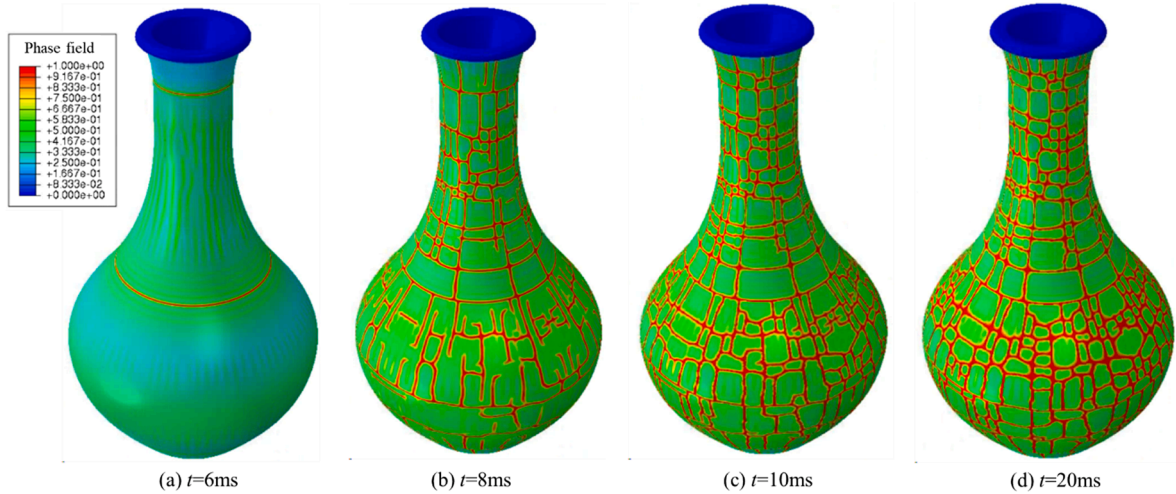


Fig. 16. Crack propagation and crack pattern formation on the surface of the ceramic bottle under rapid cooling: (a) $t = 6$ ms; (b) $t = 8$ ms; (c) $t = 10$ ms; (d) $t = 20$ ms.

of crack opening, the deformation scale factor in Fig. 17(c) and (d) is taken as 20.

Fig. 18 shows the distribution of the damage phase field on a section of the ceramic bottle. Although a large number of cracks were formed on the surface of the ceramic bottle, most of the cracks are shallow and mainly concentrated in the glaze layer as shown in Fig. 18. It should also be noted that in some locations prone to stress concentration (such as the edge of the base in Fig. 18), when the quenching time is long, it is easy to form deep cracks, even causing the breakage of the ceramic bottle. Therefore, it is necessary to accurately control the quenching time to achieve the optimal crack distribution and depth in practice.

Material parameter difference has an important influence on crack formation [51]. The CTE is a key factor affecting the pattern of rapid cooling cracks in ceramic bottles. In fact, the difference in the coefficient of thermal expansion between the inner matrix layer and the outer glaze layer is the decisive factor. Fig. 19 shows the thermal shock crack pattern formed on the surface of the ceramic bottle under different CTE differences between the inner matrix layer and the outer glaze layer. It can be found that the larger the CTE difference, the denser the crack pattern formed on the surface. Therefore, the crack pattern with a specific density can be realized by

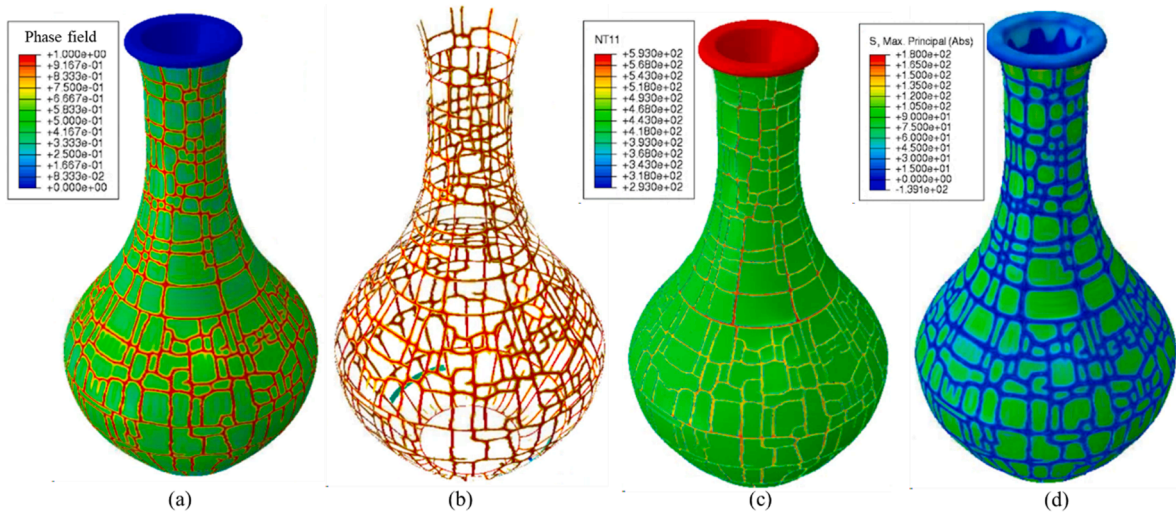


Fig. 17. Phase field simulation results of the crack pattern of the ceramic bottle under rapid cooling at 10 ms: (a) phase field distribution; (b) 3D crack topology; (c) temperature distribution; (d) maximum principal stress distribution. The deformation scale factor in (c) and (d) is 20.

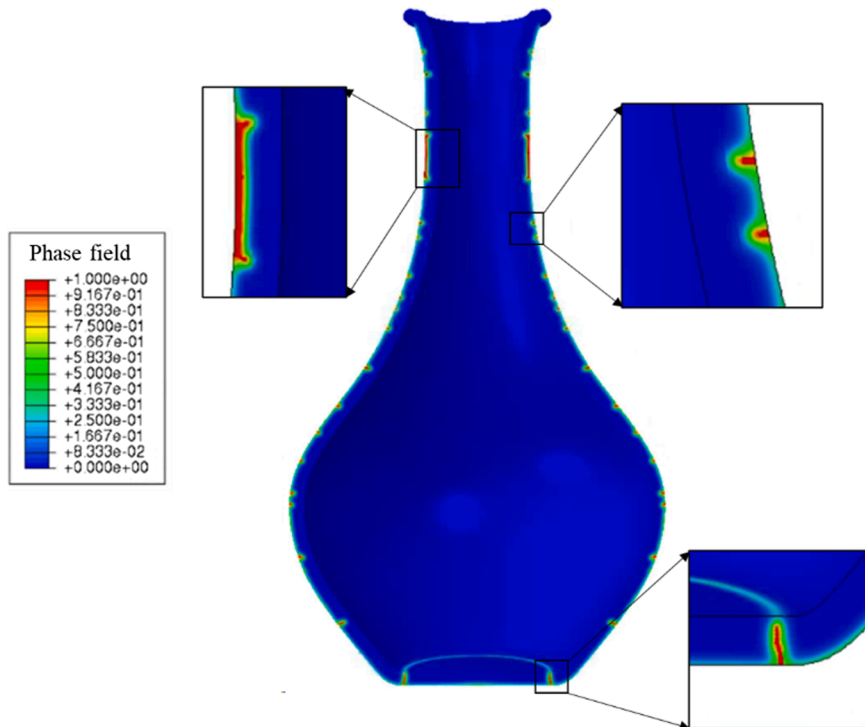


Fig. 18. Phase field simulation results of the crack pattern of the ceramic bottle under rapid cooling at 10 ms: phase field distribution on vertical section of the ceramic bottle.

controlling the CTE difference in production.

We noticed that the ice cracks on the surface of some porcelain in the museum are not entirely random but develop in a specific direction, with a unique aesthetic feeling. To reproduce this directionally crack propagation mode, we changed the cooling mode of the ceramic bottle based on the above simulation. The surface of the ceramic bottle no longer enters the cold environment completely and instantaneously, but enters the cold environment at a certain speed (4 m/s in this simulation) and in a specific direction (bottom-up and inclined 30 degrees axially compared to the ceramic bottle). Due to the asymmetry of load, we no longer use the 1/4 model, but use the 1/2 model instead. Other conditions remain unchanged. The distribution of phase field and temperature field at different times obtained by simulation is shown in Fig. 20. The overall crack growth direction is consistent with the cooling direction. It is noted that

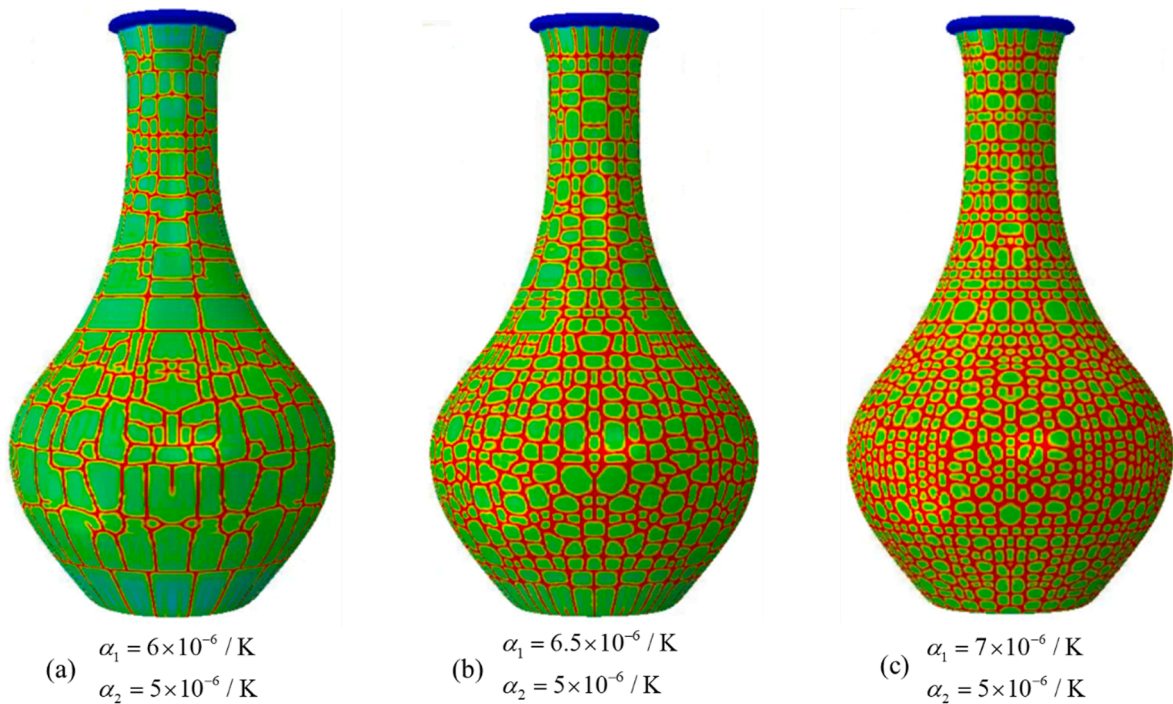


Fig. 19. Phase field simulation results of the crack pattern of the ceramic bottle under rapid cooling and various CTE differences: (a) $\Delta\alpha = 1 \times 10^{-6} / \text{K}$; (b) $\Delta\alpha = 1.5 \times 10^{-6} / \text{K}$; (c) $\Delta\alpha = 2 \times 10^{-6} / \text{K}$.

the effect of the crack “catching up” with the cooling interface is formed, but it always lags the cooling interface. These simulation results can provide valuable ideas for processing directional ice crack patterns in ceramics.

4. Conclusions

In this paper, the phase field coupled cohesive zone model is established in the commercial finite element package Abaqus/Explicit to simulate the brittle and quasi-brittle fracture processes in three-dimensional solids, especially to reproduce the complex crack patterns under thermal shock.

Several examples are given to illustrate the practicability and robustness of the model to complex fracture problems under thermal shock: from uniaxial tensile and thermal shrinkage fracture of the one-dimensional rod to nucleation, propagation, and crack pattern formation of thermal shock crack of the three-dimensional ceramic bottle. The numerical results of PF-CZM are in good agreement with the analytical solutions and previous experimental results. Based on this, the formation mechanism of crack patterns in ceramic bottles is discussed by numerical methods, especially the dominant role of the coefficient of thermal expansion difference between the ceramic matrix layer and glaze layer in forming crack patterns and the final crack density. The influence of quenching direction on crack patterns is also studied.

All examples show that the PF-CZM developed here has a good predictive capability for complex crack patterns in brittle and quasi-brittle materials, and can be used to simulate crack initiation and propagation under complex thermal shocks.

CRediT authorship contribution statement

Tao Wang: Writing – original draft, Software, Methodology, Formal analysis. **Haoyue Han:** Writing – original draft. **Yifan Wang:** Writing – original draft. **Xuan Ye:** Writing – original draft, Validation. **Guangyan Huang:** Project administration, Funding acquisition. **Zhanli Liu:** Writing – review & editing, Validation, Methodology. **Zhuo Zhuang:** Writing – review & editing, Project administration, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

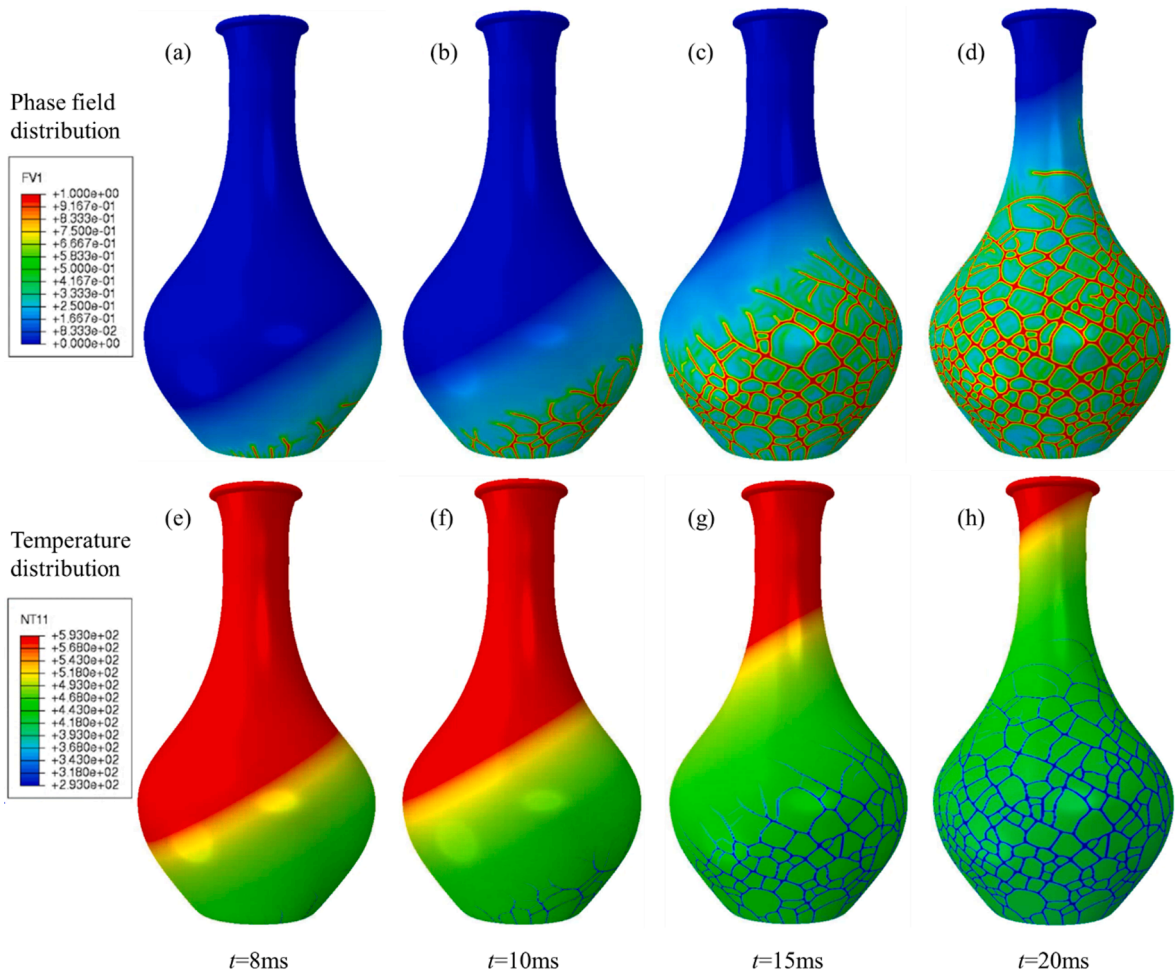


Fig. 20. Simulation results of crack phase field distribution and temperature field distribution of the ceramic bottle during directional cooling.

Data availability

No data was used for the research described in the article.

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References

- [1] Shao Y, Zhang Y, Xu X, Zhou Z, Li W, Liu B, et al. Effect of Crack Pattern on the Residual Strength of Ceramics After Quenching. *J Am Ceram Soc* 2011;94(9): 2804–7.
- [2] Bahr H-A, Weiss H-J, Bahr U, Hofmann M, Fischer G, Lampenscherf S, et al. Scaling behavior of thermal shock crack patterns and tunneling cracks driven by cooling or drying. *J Mech Phys Solids* 2010;58(9):1411–21.
- [3] Bahr H-A, Fischer G, Weiss H-J. Thermal-shock crack patterns explained by single and multiple crack propagation. *J Mater Sci* 1986;21:2716–20. <https://doi.org/10.1007/BF00551478>.
- [4] Honda S, Ogihara Y, Kishi T, Hashimoto S, Iwamoto Y. Estimation of thermal shock resistance of fine porous alumina by infrared radiation heating method. *J Ceram Soc Japan* 2009;117:1208–15. <https://doi.org/10.2109/jcersj2.117.1208>.
- [5] She J, Ohji T, Deng Z-Y. Thermal Shock Behavior of Porous Silicon Carbide Ceramics. *J Am Ceram Soc* 2002;85:2125–7. <https://doi.org/10.1111/j.1151-2916.2002.tb00418.x>.
- [6] Sadowski T, Golewski P. Cracks path growth in turbine blades with TBC under thermo – mechanical cyclic loadings. *1* 2016;10:492–9. <https://doi.org/10.3221/IGF-ESIS.35.55>.
- [7] Li J. Ice cracks in Ge-kiln porcelain. Forbidden City 1980:41.
- [8] Tarasovs S, Ghassemi A. Self-similarity and scaling of thermal shock fractures. *Phys Rev E* 2014;90:012403. <https://doi.org/10.1103/PhysRevE.90.012403>.

- [9] Li J, Song F, Jiang C. A non-local approach to crack process modeling in ceramic materials subjected to thermal shock. *Engng Fract Mech* 2015;133:85–98. <https://doi.org/10.1016/j.engfractmech.2014.11.007>.
- [10] Tang SB, Zhang H, Tang CA, Liu HY. Numerical model for the cracking behavior of heterogeneous brittle solids subjected to thermal shock. *Int J Solids Struct* 2016;80:520–31. <https://doi.org/10.1016/j.ijsolstr.2015.10.012>.
- [11] Menouillard T, Belytschko T. Analysis and computations of oscillating crack propagation in a heated strip. *Int J Fract* 2011;167:57–70. <https://doi.org/10.1007/s10704-010-9519-0>.
- [12] Dittmann M, Aldakheel F, Schulte J, Schmidt F, Krüger M, Wriggers P, et al. Phase-field modeling of porous-ductile fracture in non-linear thermo-elasto-plastic solids. *Comput Methods Appl Mech Engng* 2020;361:112730. <https://doi.org/10.1016/j.cma.2019.112730>.
- [13] Zhou S, Zhuang X, Zhu H, Rabczuk T. Phase field modelling of crack propagation, branching and coalescence in rocks. *Theor Appl Fract Mech* 2018;96:174–92. <https://doi.org/10.1016/j.tafmec.2018.04.011>.
- [14] Wu J-Y. A unified phase-field theory for the mechanics of damage and quasi-brittle failure. *J Mech Phys Solids* 2017;103:72–99. <https://doi.org/10.1016/j.jmps.2017.03.015>.
- [15] Miehe C, Schänzel L-M, Ulmer H. Phase field modeling of fracture in multi-physics problems. Part I. Balance of crack surface and failure criteria for brittle crack propagation in thermo-elastic solids. *Comput Methods Appl Mech Engng* 2015;294:449–85. <https://doi.org/10.1016/j.cma.2014.11.016>.
- [16] Wang T, Liu ZL, Cui YN, Ye X, Liu XM, Tian R, et al. A thermo-elastic-plastic phase-field model for simulating the evolution and transition of adiabatic shear band. Part I. Theory and model calibration. *Engng Fract Mech* 2020;232:107028.
- [17] Moës N, Dolbow J, Belytschko T. A finite element method for crack growth without remeshing. *Int J Numer Meth Engng* 1999;46:131–50. [https://doi.org/10.1002/\(SICI\)1097-0207\(19990910\)46:1<131::AID-NME726>3.0.CO;2-J](https://doi.org/10.1002/(SICI)1097-0207(19990910)46:1<131::AID-NME726>3.0.CO;2-J).
- [18] Song J-H, Areias PMA, Belytschko T. A method for dynamic crack and shear band propagation with phantom nodes. *Int J Numer Meth Engng* 2006;67:868–93. <https://doi.org/10.1002/nme.1652>.
- [19] Miehe C, Mauthe S. Phase field modeling of fracture in multi-physics problems. Part III. Crack driving forces in hydro-poro-elasticity and hydraulic fracturing of fluid-saturated porous media. *Comput Methods Appl Mech Engng* 2016;304:619–55. <https://doi.org/10.1016/j.cma.2015.09.021>.
- [20] Wang T, Ye X, Liu Z, Liu X, Chu D, Zhuang Z. A phase-field model of thermo-elastic coupled brittle fracture with explicit time integration. *Comput Mech* 2020; 65:1305–21. <https://doi.org/10.1007/s00466-020-01820-6>.
- [21] Molnár G, Gravouil A, Seghir R, Réthoré J. An open-source Abaqus implementation of the phase-field method to study the effect of plasticity on the instantaneous fracture toughness in dynamic crack propagation. *Comput Methods Appl Mech Engng* 2020;365:113004. <https://doi.org/10.1016/j.cma.2020.113004>.
- [22] Geelen RJM, Liu Y, Hu T, Tupek MR, Dolbow JE. A phase-field formulation for dynamic cohesive fracture. *Comput Methods Appl Mech Engng* 2019;348: 680–711.
- [23] Zhang P, Yao W, Hu X, Zhuang X. Phase field modelling of progressive failure in composites combined with cohesive element with an explicit scheme. *Compos Struct* 2021;262:113353. <https://doi.org/10.1016/j.compstruct.2020.113353>.
- [24] Li G, Yin BB, Zhang LW, Liew KM. Modeling microfracture evolution in heterogeneous composites: A coupled cohesive phase-field model. *J Mech Phys Solids* 2020;142:103968. <https://doi.org/10.1016/j.jmps.2020.103968>.
- [25] Han H, Wang T, Huang G, Liu Z, Zhuang Z. Study of spontaneous adiabatic shear bands in expanding rings under explosion by thermo-elastic-plastic phase field model. *Int J Impact Engng* 2022;161:104084. <https://doi.org/10.1016/j.ijimpeng.2021.104084>.
- [26] Zhang H, Pei X-Y, Peng H, Wu J-Y. Phase-field modeling of spontaneous shear bands in collapsing thick-walled cylinders. *Engng Fract Mech* 2021;249:107706. <https://doi.org/10.1016/j.engfractmech.2021.107706>.
- [27] Wang T, Liu ZL, Cui YN, Ye X, Liu XM, Tian R, et al. A thermo-elastic-plastic phase-field model for simulating the evolution and transition of adiabatic shear band. Part II. Dynamic collapse of thick-walled cylinder. *Engng Fract Mech* 2020;231:107027. <https://doi.org/10.1016/j.engfractmech.2020.107027>.
- [28] Wu J-Y, Mandal TK, Nguyen VP. A phase-field regularized cohesive zone model for hydrogen assisted cracking. *Comput Methods Appl Mech Engng* 2020;358: 112614. <https://doi.org/10.1016/j.cma.2019.112614>.
- [29] Martínez-Pañeda E, Golahmar A, Niordson CF. A phase field formulation for hydrogen assisted cracking. *Comput Methods Appl Mech Engng* 2018;342:742–61. <https://doi.org/10.1016/j.cma.2018.07.021>.
- [30] Heider Y, Reiche S, Siebert P, Markert B. Modeling of hydraulic fracturing using a porous-media phase-field approach with reference to experimental data. *Engng Fract Mech* 2018;202:116–34. <https://doi.org/10.1016/j.engfractmech.2018.09.010>.
- [31] Cajuhí T, Sanavia L, De Lorenzis L. Phase-field modeling of fracture in variably saturated porous media. *Comput Mech* 2018;61:299–318. <https://doi.org/10.1007/s00466-017-1459-3>.
- [32] Zhou S, Zhuang X, Rabczuk T. A phase-field modeling approach of fracture propagation in poroelastic media. *Eng Geol* 2018;240:189–203. <https://doi.org/10.1016/j.enggeo.2018.04.008>.
- [33] Wei C, Wojnar CS, Wu C. Hydro-chemo-mechanical phase field formulation for corrosion induced cracking in reinforced concrete. *Cem Concr Res* 2021;144: 106404. <https://doi.org/10.1016/j.cemconres.2021.106404>.
- [34] Cui C, Ma R, Martínez-Pañeda E. A phase field formulation for dissolution-driven stress corrosion cracking. *J Mech Phys Solids* 2021;147:104254. <https://doi.org/10.1016/j.jmps.2020.104254>.
- [35] Seles K, Aldakheel F, Tonković Z, Sorić J, Wriggers P. A general phase-field model for fatigue failure in brittle and ductile solids. *Comput Mech* 2021;67: 1431–52. <https://doi.org/10.1007/s00466-021-01996-5>.
- [36] Schreiber C, Müller R, Kuhn C. Phase field simulation of fatigue crack propagation under complex load situations. *Arch Appl Mech* 2021;91:563–77. <https://doi.org/10.1007/s00419-020-01821-0>.
- [37] Liu S, Wang Z, Zhang Y, Kou M, Bi J. The phase-field simulations of blasting failure in granites. *Int J Impact Engng* 2022;167:104274. <https://doi.org/10.1016/j.ijimpeng.2022.104274>.
- [38] Duda FP, Ciaronetti A, Toro S, Huespe AE. A phase-field model for solute-assisted brittle fracture in elastic-plastic solids. *Int J Plast* 2018;102:16–40. <https://doi.org/10.1016/j.iplas.2017.11.004>.
- [39] Chu D, Li X, Liu Z. Study the dynamic crack path in brittle material under thermal shock loading by phase field modeling. *Int J Fract* 2017;208:115–30. <https://doi.org/10.1007/s10704-017-0220-4>.
- [40] Mandal TK, Nguyen VP, Wu J-Y, Nguyen-Thanh C, de Vaucorbeil A. Fracture of thermo-elastic solids: Phase-field modeling and new results with an efficient monolithic solver. *Comput Methods Appl Mech Engng* 2021;376:113648. <https://doi.org/10.1016/j.cma.2020.113648>.
- [41] Li D, Li P, Li W, Li W, Zhou K. Three-dimensional phase-field modeling of temperature-dependent thermal shock-induced fracture in ceramic materials. *Engng Fract Mech* 2022;268:108444. <https://doi.org/10.1016/j.engfractmech.2022.108444>.
- [42] Wu X, Jiang C, Song F, Li J, Shao Y, Xu X, et al. Size effect of thermal shock crack patterns in ceramics and numerical predictions. *J Eur Ceram Soc* 2015;35(4): 1263–71.
- [43] Tanné E, Li T, Bourdin B, Marigo J-J, Maurini C. Crack nucleation in variational phase-field models of brittle fracture. *J Mech Phys Solids* 2018;110:80–99. <https://doi.org/10.1016/j.jmps.2017.09.006>.
- [44] Kumar A, Bourdin B, Francfort GA, Lopez-Pamies O. Revisiting nucleation in the phase-field approach to brittle fracture. *J Mech Phys Solids* 2020;142:104027. <https://doi.org/10.1016/j.jmps.2020.104027>.
- [45] Wu J-Y, Nguyen VP. A length scale insensitive phase-field damage model for brittle fracture. *J Mech Phys Solids* 2018;119:20–42. <https://doi.org/10.1016/j.jmps.2018.06.006>.
- [46] Ziaei-Rad V, Shen Y. Massive parallelization of the phase field formulation for crack propagation with time adaptivity. *Comput Methods Appl Mech Engng* 2016; 312:224–53. <https://doi.org/10.1016/j.cma.2016.04.013>.
- [47] Miehe C, Welschinger F, Hofacker M. Thermodynamically consistent phase-field models of fracture: Variational principles and multi-field FE implementations. *Int J Numer Meth Engng* 2010;83:1273–311. <https://doi.org/10.1002/nme.2861>.

- [48] Wu J-Y. Unified analysis of enriched finite elements for modeling cohesive cracks. *Comput Methods Appl Mech Engng* 2011;200:3031–50. <https://doi.org/10.1016/j.cma.2011.05.008>.
- [49] Xu X, Yuan W, Tian C. Dependence of Thermal Shock Crack on Specimen Width for Ceramic Materials. *Int J Appl Mechanics* 2016;08:1650042. <https://doi.org/10.1142/S1758825116500423>.
- [50] Feng X-Q, Xu M, Wang X, Gu B. Fracture mechanics analysis of three-dimensional ion cut technology. *Journal of Mechanics of Materials and Structures* 2007;2: 1831–52. <https://doi.org/10.2140/jomms.2007.2.1831>.
- [51] Gu B, Liu H-Y, Mai Y-W, Feng X-Q, Yu S-W. Fracture mechanics analysis of the effects of temperature and material mismatch on the Smart-Cut® technology. *Eng Fract Mech* 2008;75:4996–5006. <https://doi.org/10.1016/j.engfracmech.2008.06.026>.